IES · GenAI for Engineers

Lab 3.3 · Prompt
SPC Control Chart and Capability

How this prompt is built

This prompt follows **R.I.S.E.** and adds **E-MAP** for engineering context — **E**ngineering context, **M**etrics, **A**ssumptions and **P**arameters. E-MAP is what stops a generic answer: it pins the units, the assumptions that must stay visible, and the boundary of the data in scope.

	Block	What it does
R	**Role**	Sets the perspective Copilot answers from.
I	**Input**	Names exactly which evidence may be used, and forbids the rest.
S	**Steps**	Makes the work traceable so a reviewer can follow it.
E-MAP	**Engineering context, Metrics, Assumptions, Parameters**	Pins the units, the assumptions that stay visible and the data in scope.
E	**Expectations**	Defines what finished looks like, and what must never appear.
YOU	**Independent check**	Your hand calculation. This one is not Copilot's job.

Every block below is labelled with its framework letter. Keep the labels when you reuse this prompt — they are what make it repeatable.

Copy this prompt into Copilot in Excel

Select the named table first, then open Copilot and paste everything below.

R ROLE · Role

```
Act as an SPC analyst preparing a control chart review for a competent engineer.
```

I INPUT · Input

```
Use only the named Excel table SPCData. Subgroup size is n = 4. Subgroups 1 to 12 are
Baseline and subgroups 13 to 16 are Monitoring. Use the constants A2 = 0.729, D4 = 2.282
and d2 = 2.059 for n = 4. The classroom specification band is LSL 57.0 and USL 63.0 L/h.
```

S STEPS · Steps

```
1. Calculate Xbar and R for every subgroup.
2. Using the BASELINE subgroups only, calculate Xbarbar and Rbar.
3. Calculate UCL = Xbarbar + A2 x Rbar and LCL = Xbarbar - A2 x Rbar. Show both formulas.
4. Hold those baseline limits fixed and plot all 16 subgroups on one Xbar chart with the
centre line, UCL and LCL drawn.
5. Flag every monitoring subgroup that falls outside the limits.
6. Estimate sigma as Rbar divided by d2, then calculate Cp and Cpk. Show both formulas.
```

E-MAP ENGINEERING CONTEXT (E-MAP) · Engineering context, Metrics, Assumptions, Parameters

```
Engineering context: delivery test on a chemical dosing skid, measured in L/h.
Metrics: Xbar, R, control limits, Cp and Cpk, all to three decimal places.
Assumptions: the measurement system is capable and the subgroups are in time order.
State both assumptions in the output.
Parameters: n = 4, 12 baseline subgroups, limits frozen before monitoring begins.
```

E EXPECTATIONS · Expectations

```
Never recalculate the limits using the monitoring data. Report Cp and Cpk together and
explain that Cp much larger than Cpk means the process is off centre rather than incapable.
State that capability is only meaningful while the process is stable, and that these
monitoring points show it is not. A point outside a control limit is a signal to investigate,
never a diagnosis and never proof that equipment is unsafe.
End with DRAFT FOR REVIEW.
```

YOU INDEPENDENT CHECK · Independent check

```
Verify subgroup 13 by hand: Xbar = (61.5 + 61.2 + 61.4 + 61.0) / 4 = 61.275 L/h, which is
below the lower control limit. Confirm all four monitoring subgroups fall below the LCL and
that they trend steadily downwards rather than jumping once.
```

All records are fictional training data for classroom use. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use any output to operate, maintain or approve real equipment.